

Early Crohn's disease: can we change the disease course?

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Current treatments for Crohn's disease prioritize early intervention with biological therapies, with several landmark trials demonstrating that starting biologic agents within the first years of diagnosis leads to superior patient outcomes. Although challenges remain, early use of biological treatments is increasingly supported in managing moderate to severe Crohn's disease.

Crohn's disease is a chronic inflammatory bowel disease with an unpredictable course, often leading to complications such as strictures, fistulas and the need for surgery. Optimal management, especially in the early stages, has a major effect on the disease course. Efforts are being deployed to diagnose Crohn's disease in a very early disease stage in an attempt to intercept full-blown disease development. Current advancements in treatment have shifted towards highly effective strategies in the early phase of the disease, moving away from the conventional 'step-up' approach to prevent disease progression, complications and loss of work productivity.

Historically, conventional treatments for Crohn's disease have used a step-up approach, starting with corticosteroids and immunomodulators such as azathioprine, mercaptopurine and methotrexate. However, in the past decade, clinical trials have shown that early use of TNF inhibitors (infliximab (IFX) and adalimumab (ADL)) leads to better outcomes in moderate to severe cases^{1,2}. In the past few years, the early introduction of other biologic agents, such as ustekinumab and vedolizumab, was also shown to be highly effective^{3,4}.

The concept of a 'window of opportunity' in chronic inflammatory diseases such as Crohn's disease suggests that early effective treatment can prevent irreversible tissue and organ damage and hence lead to better long-term outcomes (Supplementary Table 1). Once the immune system has been fully activated and caused extensive tissue damage, it becomes more challenging to reverse or control the disease. Thus, early detection and effective treatment seem necessary to reduce the risks of complications.

Defining 'early Crohn's disease' is critical. The consensus among experts suggested that the first 2 years following diagnosis offers the best window for treatment initiation (Supplementary Table 1). However, it must be acknowledged that many patients have longstanding symptoms before a formal Crohn's disease diagnosis is made and that the 2-year period is, therefore, somewhat arbitrary. If irreversible damage is already present at diagnosis, limited surgical resection to 'reset' the inflammatory processes offers an attractive strategy in ileocolonic Crohn's disease.

A controlled trial (Dutch Trial Registry NTR1150) with 143 patients that compared laparoscopic resection with IFX in limited ileocolonic

Crohn's disease demonstrated that quality of life was comparable after 1 year (ref. 5). After 5 years, only 26% of the patients who underwent surgery had moved on to biologic agents, and 48% of the patients treated with IFX had undergone resection⁶. These results were confirmed in a larger retrospective Danish cohort study (Supplementary Table 1).

Several landmark trials have shaped our understanding of early biological therapy in Crohn's disease. Already in 2008, a study comparing step-up versus top-down induction therapy (with IFX) demonstrated that early initiation of this biological therapy led to superior outcomes¹. Schreiber and colleagues⁷ showed that clinical remission rates were highest in patients with the shortest disease duration with ADL. The CALM trial underscored that timely escalation of ADL therapy based on symptoms and biomarkers in early Crohn's disease led to superior clinical and endoscopic outcomes compared with symptom-driven escalation alone².

Two further trials with IFX offered additional evidence of the superiority of early intervention with biologic agents. In the TISKids study, children with Crohn's disease received five infusions of IFX, and in PROFILE, adult patients received IFX combined with an immunomodulator for a whole year^{8,9}. In both studies, better outcomes were achieved in the top-down compared with the accelerated step-up therapy (sustained steroid-free and surgery-free remission: 79% versus 15%, $P < 0.0001$). In PROFILE, fewer serious adverse events, including urgent abdominal surgeries, occurred in the top-down group.

In 2024, Reves and colleagues¹⁰ emphasized the importance of early intervention, demonstrating that initiating biological therapy within 12 months of diagnosis significantly (OR 3.23, $P < 0.01$) increased transmural healing rates, thereby reducing long-term complications such as fistulas and strictures.

The efficacy of early intervention with other biologic agents than anti-TNFs was investigated in several controlled trials. In the LOVE-CD trial, patients with Crohn's disease of less than 2 years and minimal previous treatment showed combined clinical and endoscopic remission at 6 and 12 months in a third of patients treated with vedolizumab, an antibody targeting $\alpha 4\beta 7$ integrin³. The gut specificity of vedolizumab is associated with very low toxicity, making it an attractive option for early intervention. Vedolizumab was also associated with lower rates of post-operative recurrence compared with placebo in the 2025-published REPREVIO study, further supporting its role in preventing disease progression (Supplementary Table 1).

Comparable outcomes were reported with ustekinumab, an antibody against the p40 subunit of IL-12 and IL-23. Although the median disease duration in the SEAVUE trial was 2.6 years, patients were naive to biologic treatment⁴. In 2025, post-hoc analyses of the phase III registration trials with guselkumab and risankizumab also reported high efficacy in 'early Crohn's disease'. Bio-naïve patients with Crohn's disease and a disease duration of <2 years had deep remission after 48 weeks of guselkumab in 45–49% of cases versus 29% with ustekinumab ($P = 0.022$) (Supplementary Table 1). Similarly, patients treated

with risankizumab reached endoscopic remission in 35% of cases for a disease duration of <2 years; 23% for 2–5 years; and 17–20% for >5 years ($P = 0.025$) (Supplementary Table 1).

Despite the clear benefits, several challenges remain in implementing early biological therapy for eligible patients. Diagnostic delays are a major obstacle, as the ‘clock’ is already ticking by the time most patients are diagnosed with Crohn’s disease (Supplementary Table 1). Emerging strategies, such as targeting Crohn’s disease in the preclinical phase, are being explored to intercept the disease even before substantial inflammation occurs. As the field continues to evolve, future research is likely to focus on identifying biomarkers that can predict which patients will benefit the most from early intervention and optimizing therapeutic strategies to maintain remission and prevent long-term complications. Lastly, it is important to acknowledge that many biological therapies are still not optimally available or affordable in developing countries, highlighting the need for initiatives to improve availability and affordability.

In conclusion, early intervention with biological therapy offers a promising strategy to alter the natural, progressive course of Crohn’s disease. Although challenges such as diagnostic delays and patient access to early therapy remain, the growing body of evidence supports the use of biologic agents as a first-line treatment in moderate to severe Crohn’s disease. Going forwards, efforts to refine treatment algorithms and enhance early diagnosis will be key to diminishing the global burden of Crohn’s disease.

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Competing interests

The authors declare no competing interests.

Additional information

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